



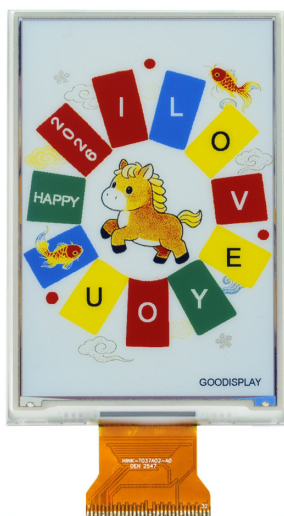
# **3.7 inch E-paper Display Series**



**GDEH037E01**

Dalian Good Display Co., Ltd.

# Product Specifications



|             |                       |
|-------------|-----------------------|
| Customer    | Standard              |
| Description | ePaper Display Series |
| Model Name  | GDEH037E01            |
| Date        | 2026/03/20            |
| Revision    | 1.0                   |

|  | Design Engineering                                                                  |                                                                                       |                                                                                       |
|--|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  | Approval                                                                            | Check                                                                                 | Design                                                                                |
|  |  |  |  |

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## 1. Overview

GDEH037E01 is an Active Matrix E Ink Spectra 6 Display, with interface and a reference system design. The 3.7" active area contains 480×720 pixels, and has 3-bit B/W/R/Y/B/G full display capabilities. An integrated circuit contains gate buffer, source buffer, interface, timing control logic, oscillator, DC-DC, SRAM, LUT, VCOM and border are supplied with each panel.

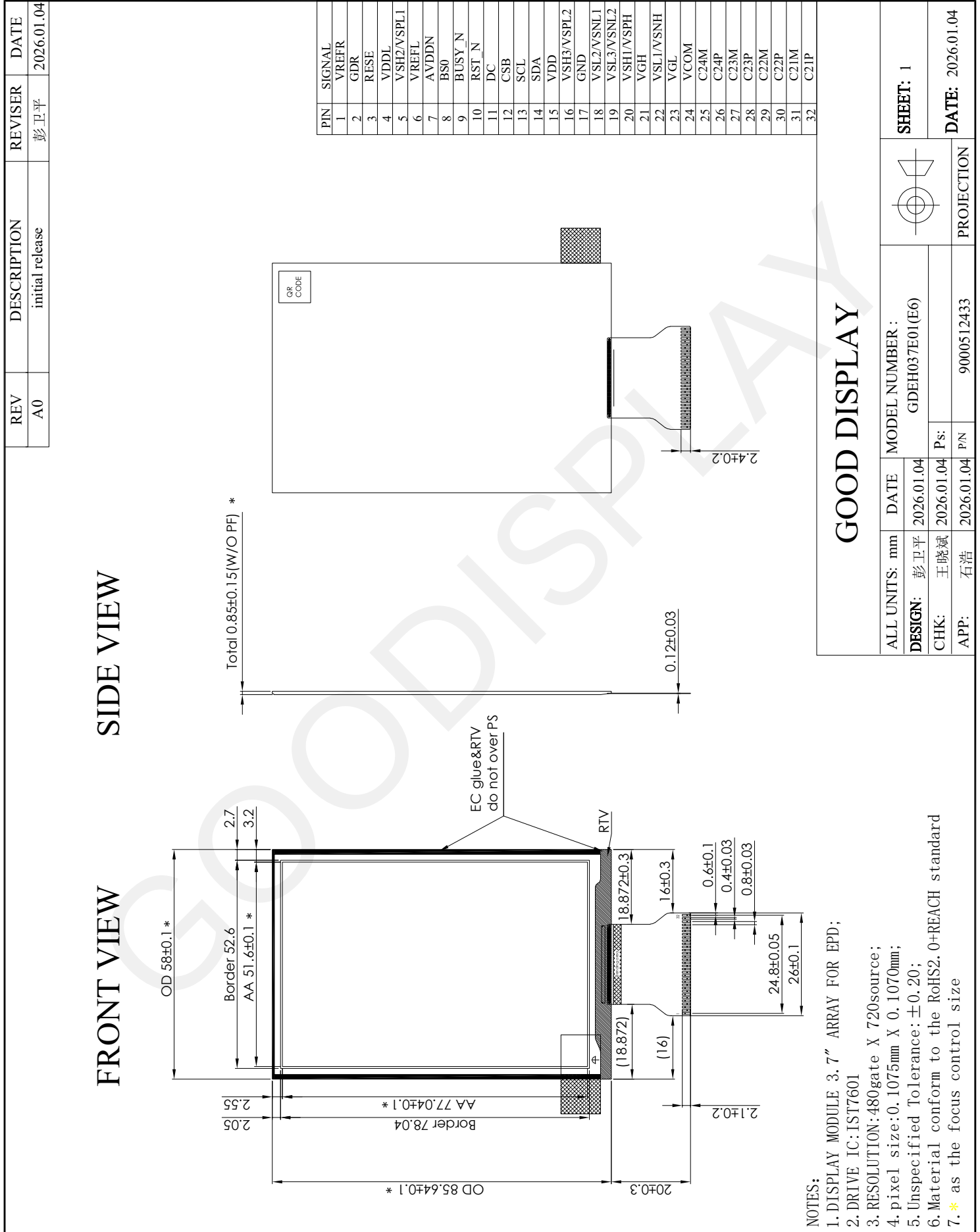
## 2. Features

- 480\*720 pixels display
- High contrast High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable display
- Commercial temperature range
- Landscape, portrait modes
- Hard-coat antiglare display surface
- Ultra Low current deep sleep mode
- On chip display RAM
- Low voltage detect for supply voltage
- High voltage ready detect for driving voltage
- Internal temperature sensor
- Waveform stored in On-chip OTP
- Serial peripheral interface available
- On-chip oscillator
- On-chip booster and regulator control for generating VCOM, Gate and Source driving voltage
- I2C signal master interface to read external temperature sensor/  
built-in temperature sensor

### 3. Mechanical Specifications

| Parameter           | Specifications          | Unit  | Remark                      |
|---------------------|-------------------------|-------|-----------------------------|
| Screen Size         | 3.7                     | Inch  |                             |
| Display Resolution  | 480(H)×720(V)           | Pixel |                             |
| Active Area         | 50.40 × 75.60           | mm    |                             |
| Pixel Pitch         | 0.105 × 0.105           | mm    |                             |
| Pixel Configuration | Rectangle               |       |                             |
| Outline Dimension   | 58(H)×85.64 (V) ×0.9(D) | mm    | <b>Without masking film</b> |
| IC                  | TBD                     |       |                             |
| Weight              | 8.5±0.5                 | g     |                             |

## 4. Mechanical Drawing of EPD module



### NOTES:

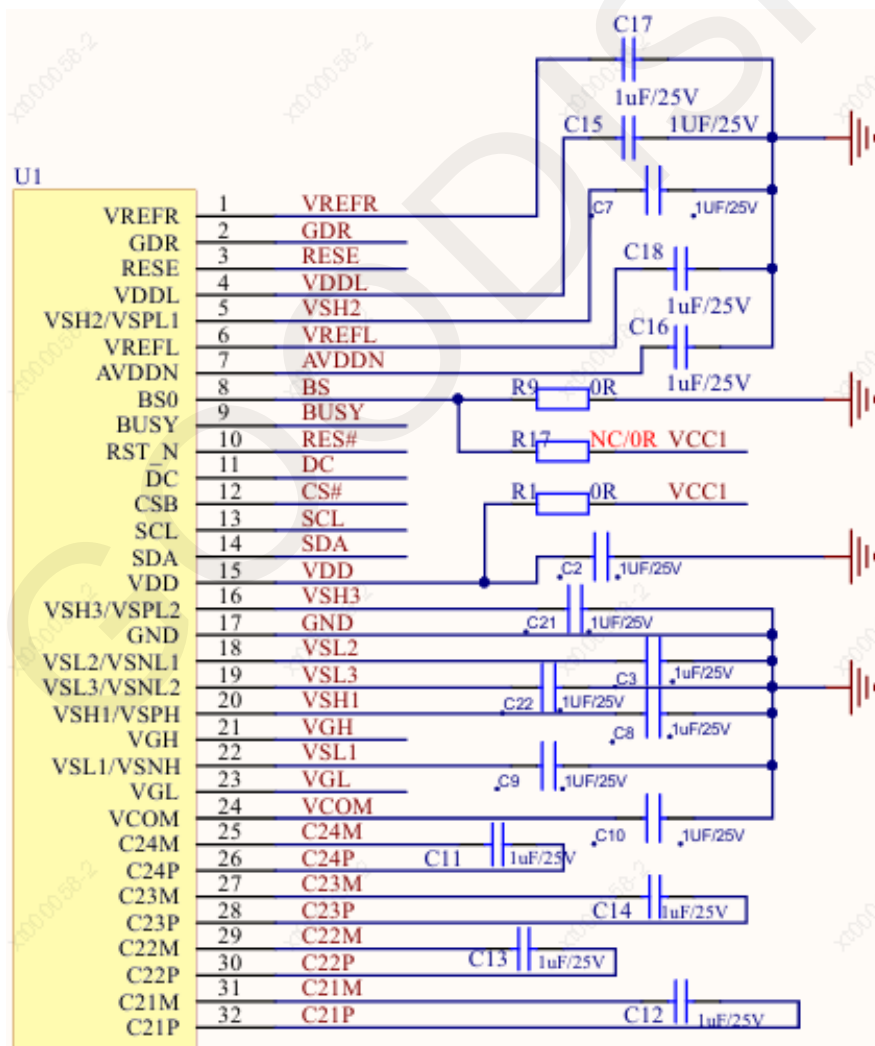
1. DISPLAY MODULE 3.7" ARRAY FOR EPD;
2. DRIVE IC: IST7601
3. RESOLUTION: 480gate X 720source;
4. pixel size: 0.1075mm X 0.1070mm;
5. Unspecified Tolerance: ±0.20;
6. Material conform to the RoHS2.0+REACH standard
7. \* as the focus control size

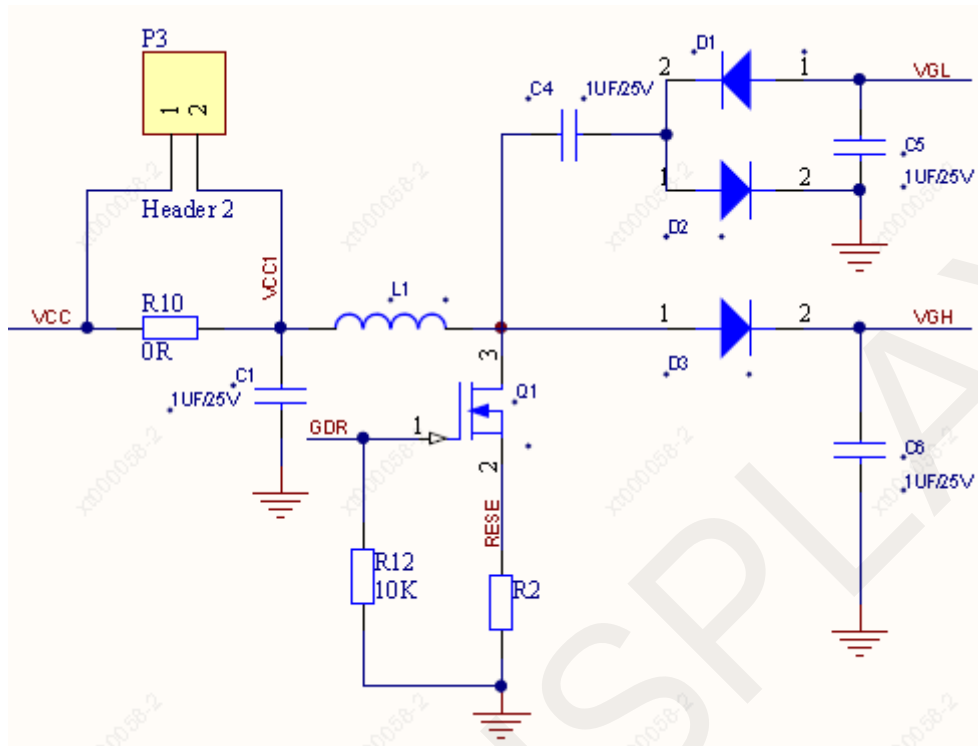
## 5. Input/output Terminals

| Pin # | Single     | Description                                                                                       | Remark   |
|-------|------------|---------------------------------------------------------------------------------------------------|----------|
| 1     | VREFR      | Digital power                                                                                     |          |
| 2     | GDR        | N-MOS gate control, for booster circuit output.                                                   |          |
| 3     | RESE       | Current sense input for control loop                                                              |          |
| 4     | VDDL       | Digital power                                                                                     |          |
| 5     | VSH2/VSPL1 | Positive source driver Voltage                                                                    |          |
| 6     | VREFL      | Digital power                                                                                     |          |
| 7     | AVDDN      | Charge Pump voltage                                                                               |          |
| 8     | BS0        | Bus Selection. Select 3-wire / 4-wire SPI interface.<br>L: 4-wire interface. H: 3-wire interface. | Note 6-5 |
| 9     | BUSY       | Driver busy flag. L: Driver is Busy. H: Host side can send command/data to driver                 | Note 6-4 |
| 10    | RST_N      | Global reset pin                                                                                  | Note 6-3 |
| 11    | DC         | Serial communication Command/Data input                                                           | Note 6-2 |
| 12    | CSB        | Serial communication chip select.                                                                 | Note 6-1 |
| 13    | SCL        | Serial communication clock input                                                                  |          |
| 14    | SDA        | Serial communication data input/output                                                            |          |
| 15    | VDD        | Digital power                                                                                     |          |
| 16    | VSH3/VSPL2 | Positive source driver Voltage                                                                    |          |
| 17    | GND        |                                                                                                   |          |
| 18    | VSL2/VSNL1 | Negative source driver voltage                                                                    |          |
| 19    | VSL3/VSNL2 | Negative source driver voltage                                                                    |          |
| 20    | VSH1/VSPH  | Positive source driver Voltage                                                                    |          |
| 21    | VGH        | Positive Gate driving voltage                                                                     |          |
| 22    | VSL1/VSNH  | Negative source driver voltage                                                                    |          |
| 23    | VGL        | Negative Gate voltage.                                                                            |          |
| 24    | VCOM       | VCOM output                                                                                       |          |
| 25    | C24M       | Digital power                                                                                     |          |
| 26    | C24P       | Digital power                                                                                     |          |
| 27    | C23M       | Digital power                                                                                     |          |
| 28    | C23P       | Digital power                                                                                     |          |
| 29    | C22M       | Digital power                                                                                     |          |
| 30    | C22P       | Digital power                                                                                     |          |
| 31    | C21M       | Digital power                                                                                     |          |
| 32    | C21P       | Digital power                                                                                     |          |

1. Note 6-1: This pin (CSB) is the chip select input connecting to the MCU. The chip is enabled for MCU communication: only when CSB is pulled LOW.
2. Note 6-2: This pin (DC) is Data/Command control pin connecting to the MCU. When the pin is pulled HIGH, the data will be interpreted as data. When the pin is pulled LOW, the data will be interpreted as command.
3. Note 6-3: This pin (RST\_N) is reset signal input. The Reset is active low.
4. Note 6-4: This pin (BUSY\_N) is busy state output pin. When Busy is High, the operation of chip should not be interrupted and any commands should not be issued to the module. The driver IC will put Busy pin High when the driver IC is working such as: Outputting display waveform; Communicating with digital temperature sensor
5. Note 6-5: This pin (BS) is for 3-line SPI or 4-line SPI selection. When it is “Low”, 4-line SPI is selected. When it is “High”, 3-line SPI (9 bits SPI) is selected.

## 6. Reference Circuit





| Part Name           | Value   | IST7163 /Reference Part                                                                                                                                                                         |
|---------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1—C18 ;<br>C21-C22 | 1uF     | 0603;X5R;Voltage Rating: 25V                                                                                                                                                                    |
| D1—D3               | MBR0530 | MBR0530<br>1) Reverse DC voltage $\geq$ 30V<br>2) $I_o \geq 500\text{mA}$<br>3) Forward voltage $\leq$ 430mV                                                                                    |
| R2                  | 2.2ohm  | 2.2Ω/0603/1% variation                                                                                                                                                                          |
| Q1                  | MOS     | Si1304BDL/NX3008NBK:<br>1) Drain-Source breakdown voltage $\geq 30\text{V}$<br>2) $V_{gs(th)} = 0.9\text{V (Typ), } 1.3\text{V (Max)}$<br>3) $R_{ds on} \leq 2.1 \Omega @ V_{gs} = 2.5\text{V}$ |
| L1                  | 47uH    | NRH3010T100MN/NRH3010T470MN                                                                                                                                                                     |

## 7. Absolute Maximum Rating

| Symbol           | Parameter                        | Rating       | Unit | Humidity | Unit | Note                          |
|------------------|----------------------------------|--------------|------|----------|------|-------------------------------|
| V <sub>DD</sub>  | Logic supply voltage             | -0.3 to +6.0 | V    | -        | -    |                               |
| T <sub>OPR</sub> | Operation temperature range      | 0 to 50      | °C   | 45 to 70 | %    |                               |
| T <sub>ttg</sub> | Transportation temperature range | -25 to 50    | °C   | 45 to 70 | -    | Note 11-2                     |
| T <sub>stg</sub> | Storage condition                | -25 to 50    | °C   | 45 to 70 | %    | Maximum storage time: 5 years |

Note 7-1: We guarantee the single pixel display quality for 0-35°C, but we only guarantee the barcode readable for 35-40°C. Normal use is recommended to refresh every 24 hours.

Note 7-2: Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Electrical Characteristics chapter.

Note 7-3: T<sub>ttg</sub> is the transportation condition, the transport time is within 10 days for -25°C ~ 0°C or 50°C ~ 60°C.

Note 7-4: When the four-color product is stored. The display screen should be kept white and face up. We suggest that the full black and full white picture could be added to clear the screen after the module is refreshed for a long time, the display effect would be better.

Note 7-5: Due to the nature of FPL, we will not control the display effect of yellow and white characters on white and yellow characters on white.

## 8. DC Characteristics

The following specifications apply for: V<sub>SS</sub>=0V, V<sub>DD</sub>=3.0V, T<sub>OPR</sub>=25°C

| Symbol              | Parameter                 | Test Condition           | Min.                 | Typ. | Max.                | Unit |
|---------------------|---------------------------|--------------------------|----------------------|------|---------------------|------|
| V <sub>DD</sub>     | VDD operation voltage     | -                        | 2.4                  | 3.0  | 3.6                 | V    |
| V <sub>IH</sub>     | High level input voltage  | -                        | 0.7xV <sub>IO</sub>  | -    | V <sub>IO</sub>     | V    |
| V <sub>IL</sub>     | Low level input voltage   | -                        | GND                  | -    | 0.3xV <sub>DD</sub> | V    |
| V <sub>OH</sub>     | High level output voltage | I <sub>OH</sub> = 400μA  | V <sub>IO</sub> -0.4 | -    | -                   | V    |
| V <sub>OL</sub>     | Low level output voltage  | I <sub>OL</sub> = -400μA | GND                  | -    | GND+0.4             | V    |
| I <sub>update</sub> | Module operating current  | -                        | -                    | TBD  | -                   | mA   |
| I <sub>sleep</sub>  | Deep sleep mode           | V <sub>CI</sub> =3.3V    | -                    | TBD  | -                   | uA   |

The Typical power consumption is measured using associated 25°C waveform with following pattern transition: from horizontal scan pattern to vertical scan pattern.

The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by Gooddisplay.

## 9. Power Consumption (Price tag screen)

| Parameter                             | Symbol | Conditions | TYP | Max | Unit | Remark |
|---------------------------------------|--------|------------|-----|-----|------|--------|
| Panel power consumption during update | -      | 25°C       | TBD |     | mAs  | -      |
| Deep sleep mode                       | -      | 25°C       | TBD |     | uA   | -      |

mAs=update average current×update time

## 10. Optical characteristics

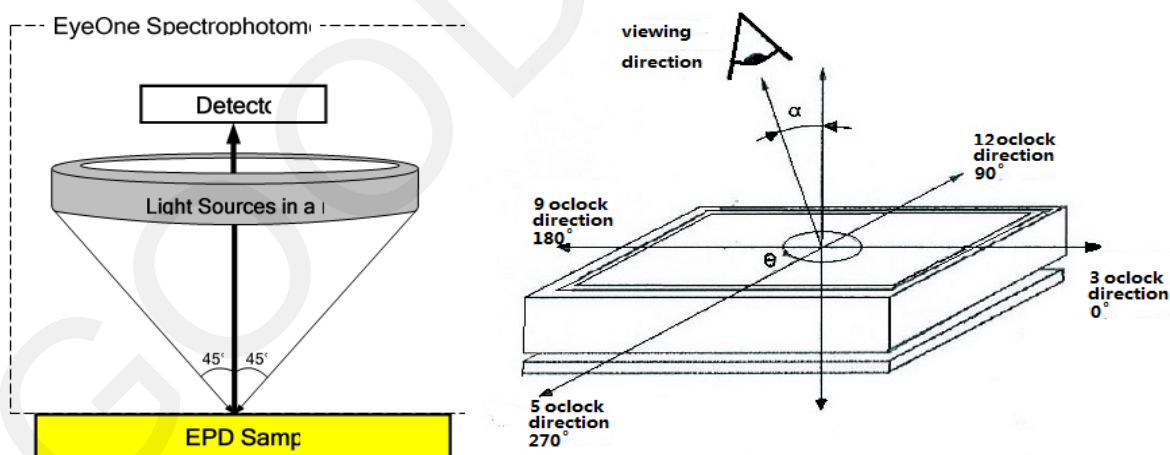
### 10.1 Optical Measurement Conditions

| Item                | Symbol     | Value    | Unit | Note           |
|---------------------|------------|----------|------|----------------|
| Ambient Temperature | Ta         | 25±2     | °C   | Indoor testing |
| Ambient Humidity    | Ha         | 50±5     | %RH  | -              |
| Supply Voltage      | VCC, VDDIO | 3.0      | V    | -              |
| illuminance         | -          | 800~1300 | Lux  |                |

Note 10-1: Image is updated with above condition

### 10.2 Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area (Rl) and the reflectance in a dark area (Rd):  $CR = Rl/Rd$



### 10.3 Reflection Ratio

The reflection ratio is expressed as:

$R = \text{Reflectance Factor white board} \times (L_{\text{center}} / L_{\text{white board}})$

$L_{\text{center}}$  is the luminance measured at center in a white area ( $R=G=B=1$ ).  $L_{\text{white board}}$  is the luminance of a standard white board.

Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.

## 11. Handling Safety And Environmental Requirements

### WARNING

The display module should be kept flat or fixed to a rigid, curved support with limited bending along the long axis. It should not be used for continual flexing and bending. Handle with care. Should the display break do not touch any material that leaks out. In case of contact with the leaked material then wash with water and soap.

### CAUTION

The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components.

Disassembling the display module can cause permanent damage and invalidate the warranty agreements.

IPA solvent can only be applied on active area and the back of a glass. For the rest part, it is not allowed.

Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricity and other rough environmental conditions.

### Mounting Precautions

(1) It's recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module.

(2) It's recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force.

(3) You should adopt radiation structure to satisfy the temperature specification.

(4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction.

(5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS)

(6) When there is dust on the surface, gently wipe it with absorbent cotton or a dust-free cloth. It is recommended to use anhydrous ethanol to clean the adhesive used to attach PS.

(7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading.

### Data sheet status

Product specification

The data sheet contains final product specifications.

### Limiting values

Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.

### Application information

Where application information is given, it is advisory and does not form part of the specification.

| Product Environmental certification                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROHS                                                                                                                                                                                                                                                                                |
| REMARK                                                                                                                                                                                                                                                                              |
| All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation. |

## 12. Reliability test

### 12.1 Reliability test items

|    | TEST                                      | CONDITION                                                                                                           | REMARK                   |
|----|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1  | High-Temperature Operation                | T=40°C, RH=35%RH, For 240Hrs                                                                                        |                          |
| 2  | Low-Temperature Operation                 | T = 0°C for 240 hrs                                                                                                 |                          |
| 3  | High-Temperature Storage                  | T=50°C RH=35%RH For 240Hrs                                                                                          | Test in white pattern    |
| 4  | Low-Temperature Storage                   | T = -25°C for 240 hrs                                                                                               | Test in white pattern    |
| 5  | High Temperature, High-Humidity Operation | T=40°C, RH=90%RH, For 168Hrs                                                                                        |                          |
| 6  | High Temperature, High-Humidity Storage   | T=50°C, RH=90%RH, For 240Hrs                                                                                        | Test in white pattern    |
| 7  | Temperature Cycle                         | -25°C(30min)~60°C(30min), 100 Cycle                                                                                 | Test in white pattern    |
| 8  | Package Vibration                         | 1.04G,Frequency : 20~200Hz<br>Direction : X,Y,Z<br>Duration: 30 minutes in each direction                           | Full packed for shipment |
| 9  | Package Drop Impact                       | Drop from height of 100 cm on<br>Concrete surface<br>Drop sequence:1 corner, 3edges,<br>6face<br>One drop for each. | Full packed for shipment |
| 10 | UV exposure Resistance                    | 765 W/m <sup>2</sup> for 168hrs,40°C                                                                                |                          |
| 11 | Electrostatic Discharge(no-operation)     | (Machine model) :+/-200v, 0Ω,200PF                                                                                  |                          |

Actual EMC level to be measured on customer application. Note1: Stay white pattern for storage and non-operation test. Note2: Operation is black/white pattern , hold time is 150S. Note3: Keep testing after 2 hours placing at 20°C-25°C.

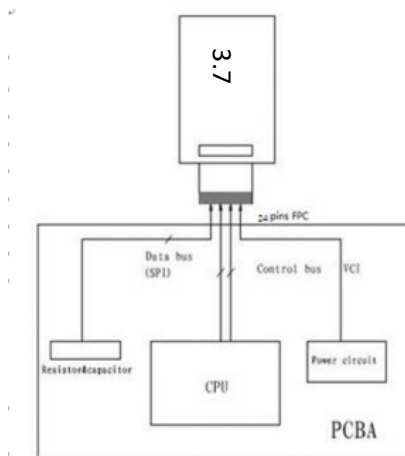
## 12.2 Product life time

The EPD Module is designed for a 5 -year life-time with 20 °C/60%RH operation assumption. Reliability estimation testing with accelerated life-time theory would be demonstrated to provide confidence of EPD lifetime.

## 12.3 Product warranty

Warranty conditions have to be negotiated between GOODISPLAY and individual customers. GOODISPLAY provides 12+1 (one month delivery time) months warranty for all products which are purchased from GOODISPLAY.

## 13. Block Diagram



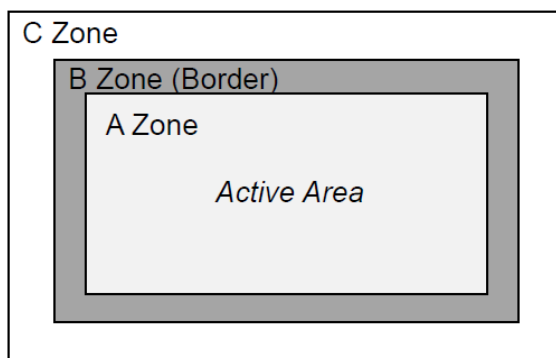
## 14. Shipment inspection specification

### 14.1 Zone Definition

A Zone: Active Area

B Zone: Border Area

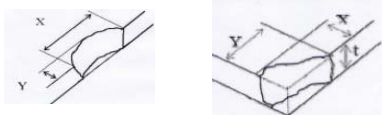
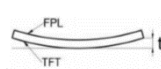
C Zone: From B Zone edge to panel edge



## 14.2 Line/Spot defect size



## 14.3 Point and line standard

| Shipment Inspection Standard                    |                                                          |                                                                                      |             |                                                                                       |        |             |
|-------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------|--------|-------------|
| Equipment: Electrical test fixture, Point gauge |                                                          |                                                                                      |             |                                                                                       |        |             |
| Outline dimension                               | 58(H)×85.64 (V)<br>×0.9(D)                               | Unit: mm                                                                             | Part-A      | Active area                                                                           | Part-B | Border area |
| Environment                                     | Temperature                                              | Humidity                                                                             | illuminance | Distance                                                                              | Time   | Angle       |
|                                                 | 19℃～25℃                                                  | 50±5%RH                                                                              | 800~1300Lux | 300 mm                                                                                | 35Sec  | 45°         |
| Defect type                                     | Inspection method                                        | Standard                                                                             |             | Part-A                                                                                | Part-B |             |
| Spot                                            | Electric Display                                         | D≤0.25 mm                                                                            |             | Ignore                                                                                | Ignore |             |
|                                                 |                                                          | 0.25 mm<D≤0.4 mm                                                                     |             | N≤4                                                                                   | Ignore |             |
|                                                 |                                                          | D>0.4 mm                                                                             |             | Not Allow                                                                             | Ignore |             |
| Display malfunction                             | Electric Display                                         | Not Allow                                                                            |             | Not Allow                                                                             | Ignore |             |
| Display error                                   | Electric Display                                         | Not Allow                                                                            |             | Not Allow                                                                             | Ignore |             |
| Scratch or line defect(include dirt)            | Visual/Film card                                         | L≤2 mm,W≤0.2 mm                                                                      |             | Ignore                                                                                | Ignore |             |
|                                                 |                                                          | 2.0mm<L≤5.0mm,0.2<W≤0.3mm,                                                           |             | N≤2                                                                                   | Ignore |             |
|                                                 |                                                          | L>5 mm,W>0.3 mm                                                                      |             | Not Allow                                                                             | Ignore |             |
| PS Bubble                                       | Visual/Film card                                         | D≤0.2mm                                                                              |             | Ignore                                                                                | Ignore |             |
|                                                 |                                                          | 0.2mm≤D≤0.35mm                                                                       |             | N≤4                                                                                   | Ignore |             |
|                                                 |                                                          | D>0.35 mm                                                                            |             | Not Allow                                                                             | Ignore |             |
| Corner /Edge chipping                           | Visual/Film card                                         | X≤6mm,Y≤0.4mm, Do not affect the electrode circuit (Edge chipping)                   |             |                                                                                       |        |             |
|                                                 |                                                          | X≤1mm,Y≤1mm, Do not affect the electrode circuit( (Corner chipping)                  |             |                                                                                       |        |             |
|                                                 |                                                          | Ignore                                                                               |             |                                                                                       |        |             |
|                                                 |                                                          |  |             |                                                                                       |        |             |
| TFT warping                                     | For 1.54~7.5inch, T≤2mm; For above 7.5inch, T≤3mm        |                                                                                      |             |  |        |             |
| Remark                                          | 1.Cannot be defect & failure cause by appearance defect; |                                                                                      |             |                                                                                       |        |             |
|                                                 | 2.Cannot be larger size cause by appearance defect;      |                                                                                      |             |                                                                                       |        |             |
|                                                 | L=long      W=wide      D=point size      N=Defects NO   |                                                                                      |             |                                                                                       |        |             |

## 15. Packing

Full carton: 12 pcs tray with products and 1 pcs empty tray.

Last carton: less 12pcs tray with products and 1 pcs empty tray.

The packager will add the right amount of EPE to box so that the added EPE 's height approximately close to the EPE height around the inside of the box.

